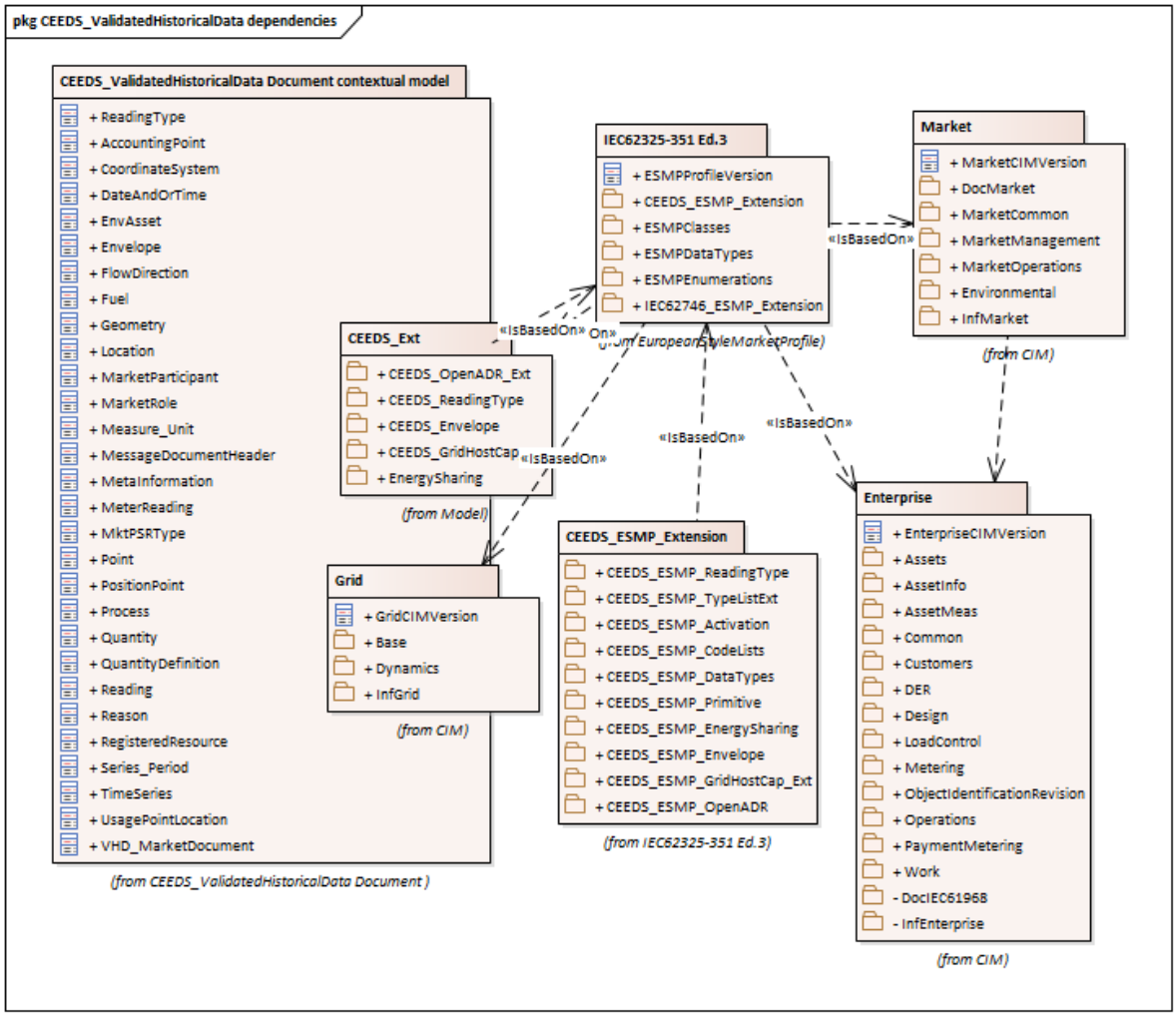
CEEDS_ValidatedHistoricalData Document contextual model

Overview of the model

!# shows the model.



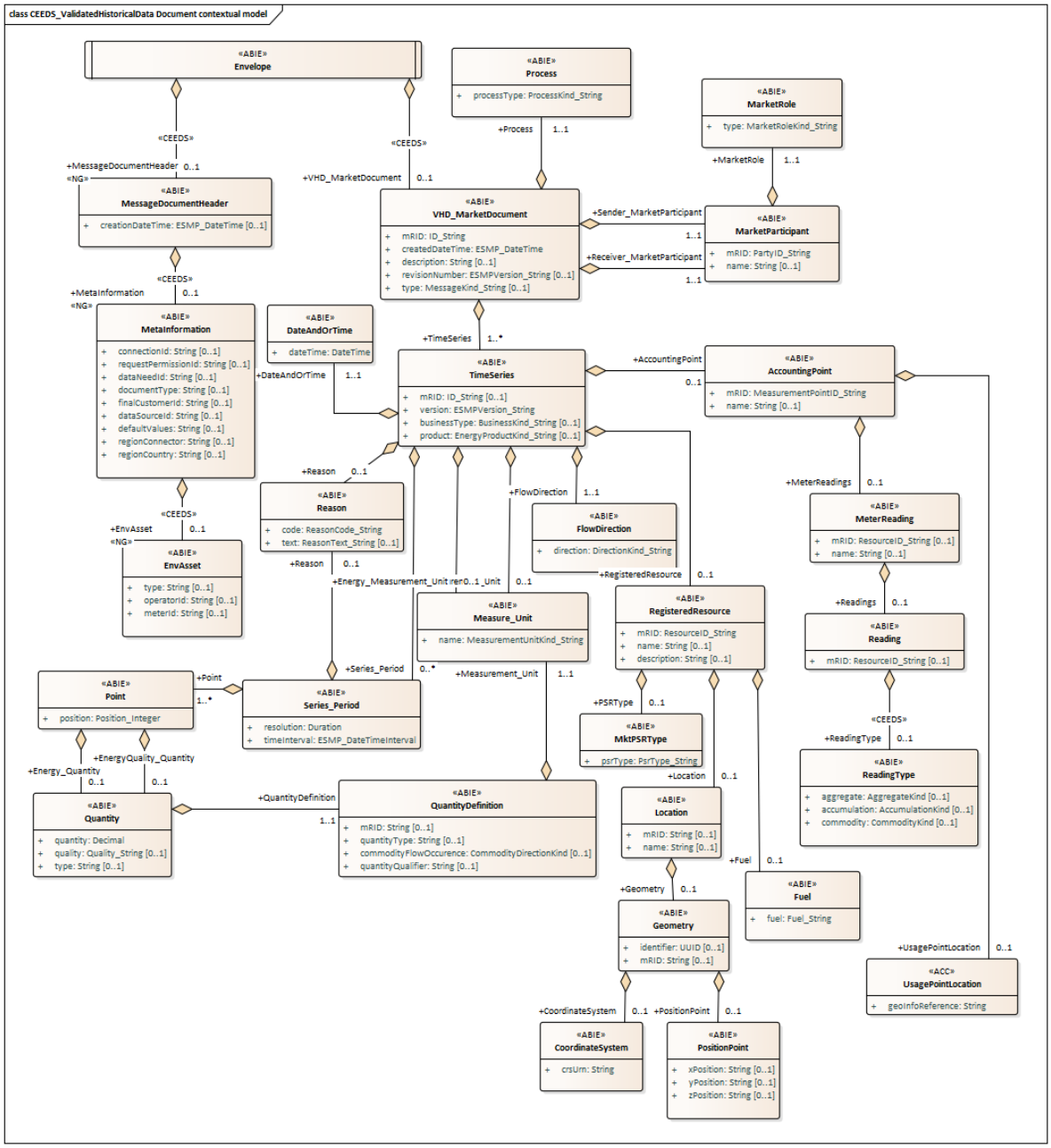


Figure !# - CEEDS_ValidatedHistoricalData Document contextual model

IsBasedOn relationships from the European style market profile

!# shows the traceability dependency of the classes used in this package towards the upper level.

Table !# - IsBasedOn dependency

Name	Complete IsBasedOn Path
AccountingPoint	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::AccountingPoint
CoordinateSystem	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::CoordinateSystem
DateAndOrTime	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::DateAndOrTime
EnvAsset	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::EnvAsset

Envelope	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::Envelope
FlowDirection	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::FlowDirection
Fuel	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::Fuel
Geometry	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::Geometry
Location	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::Location
MarketParticipant	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::MarketParticipant
MarketRole	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::MarketRole
Measure_Unit	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::Measure_Unit
MessageDocumentHeader	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::MessageDocumentHeader
MetaInformation	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::MetaInformation
MeterReading	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::MeterReading
MktPSRType	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::MktPSRType
Point	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::Point
PositionPoint	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::PositionPoint
Process	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::Process
Quantity	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::Quantity
QuantityDefinition	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::QuantityDefinition
Reading	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::Reading
ReadingType	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::ReadingType
Reason	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::Reason
RegisteredResource	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::RegisteredResource
Series_Period	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData

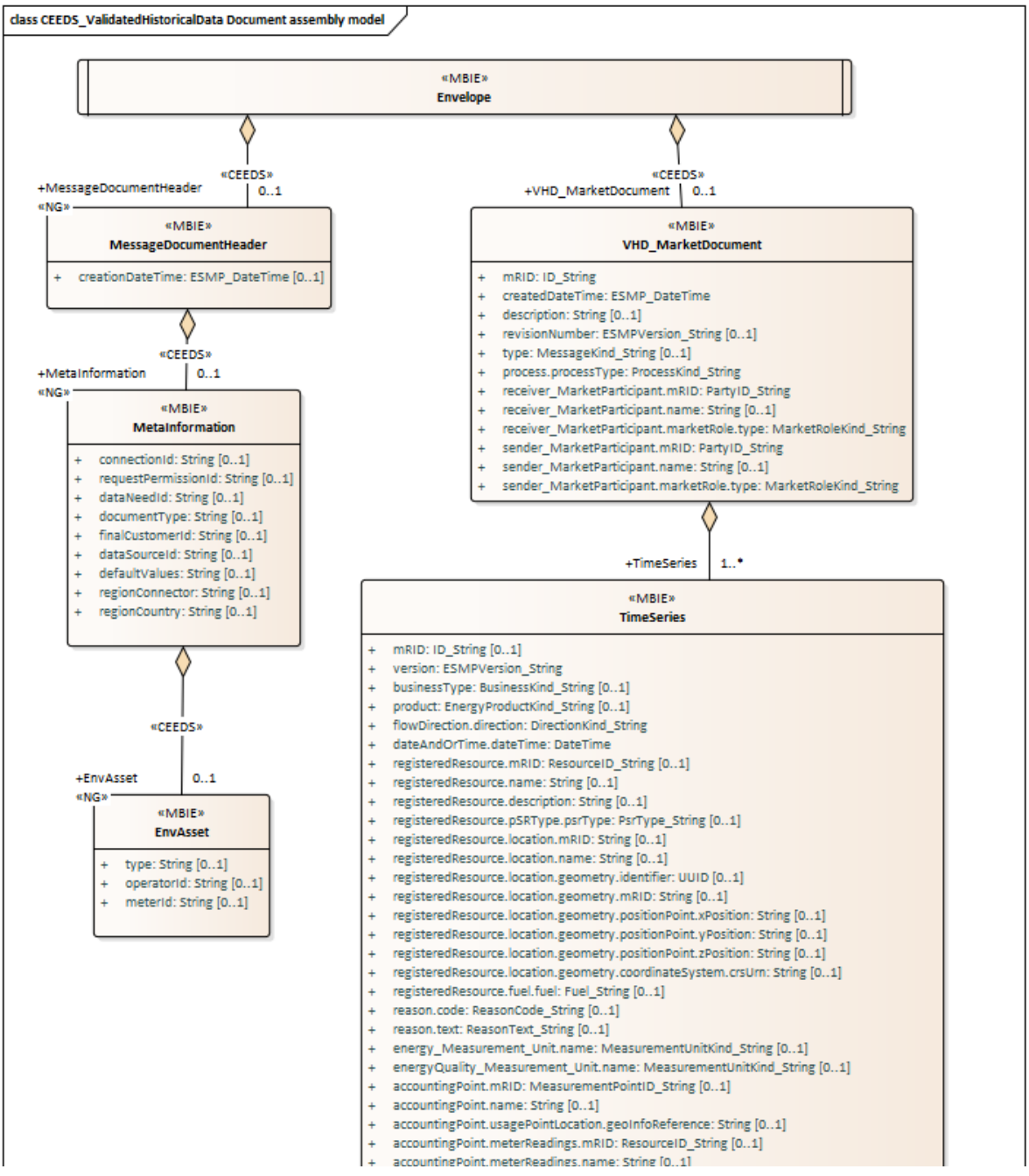
	Document ::CEEDS_ValidatedHistoricalData Document contextual model::Series_Period
TimeSeries	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::TimeSeries
UsagePointLocation	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::UsagePointLocation
VHD_MarketDocument	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document contextual model::VHD_MarketDocument

##MXM##

CEEDS_ValidatedHistoricalData Document assembly model

Overview of the model

!# shows the model.



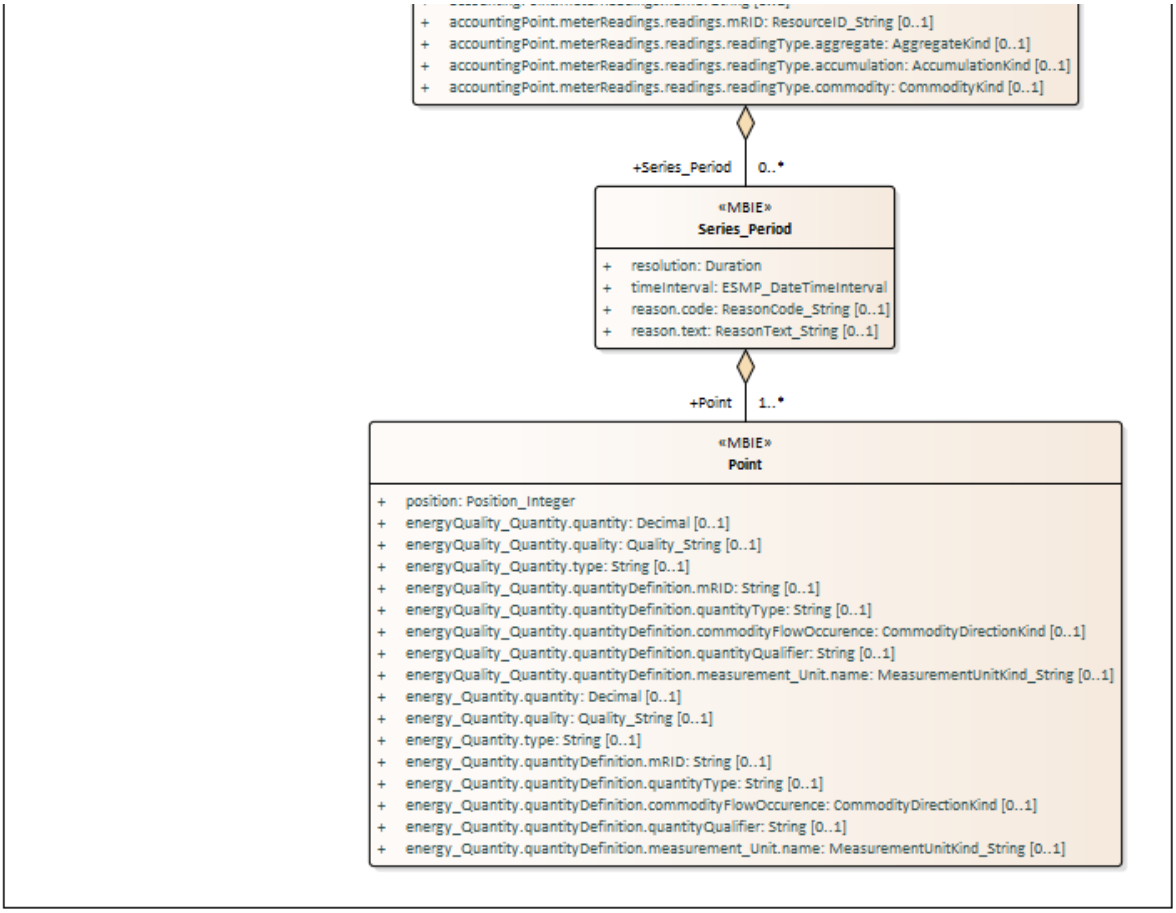


Figure !# - CEEDS_ValidatedHistoricalData Document assembly model

IsBasedOn relationships from the European style market profile

!# shows the traceability dependency of the classes used in this package towards the upper level.

Table !# - IsBasedOn dependency

Name	Complete IsBasedOn Path
EnvAsset	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document assembly model::EnvAsset
Envelope	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document assembly model::Envelope
MessageDocumentHeader	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document assembly model::MessageDocumentHeader
MetaInformation	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document assembly model::MetaInformation
Point	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document assembly model::Point
Series_Period	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document assembly model::Series_Period
TimeSeries	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document assembly model::TimeSeries
VHD_MarketDocument	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_ValidatedHistoricalData Document ::CEEDS_ValidatedHistoricalData Document assembly model::VHD_MarketDocument

##MXM##

Detailed CEEDS_ValidatedHistoricalData Document assembly model

Envelope root class

The envelope entity is a non-CIM object used in an energy data space environment to serve as a central top-level entity connecting the MarketDocumentHeader (energy data space specific context information block) with the regular CIM MarketDocument.

!# shows all association ends of Envelope with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document assembly model::Envelope with other classes

Order	mult.	Class name / Role	Description
0	[0..1]	MessageDocumentHeader MessageDocumentHeader	Association Based On : CEEDS_ValidatedHistoricalData Document contextual model::Envelope.[] ----- CEEDS_ValidatedHistoricalData Document contextual model::MessageDocumentHeader.MessageDocumentHeader[0..1]
1	[0..1]	VHD_MarketDocument VHD_MarketDocument	Association Based On : CEEDS_ValidatedHistoricalData Document contextual model::Envelope.[] ----- CEEDS_ValidatedHistoricalData Document contextual model::VHD_MarketDocument.VHD_MarketDocument[0..1]

##MXM##

EnvAsset

EnvAsset is an energy data space specific attribute referring to the Asset from which data is being requested

!# shows all attributes of EnvAsset.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document assembly model::EnvAsset

Order	mult.	Attribute name / Attribute type	Description
0	[0..1]	type String	type is an energy data space specific attribute that specifies type of asset refered to.
1	[0..1]	operatorId String	operatorId is an energy data space specific attribute that specifies the id operator of the asset refered to.
2	[0..1]	meterId String	meterId is an energy data space specific attribute that specifies the id of the meter that measures the asset.

##MXM##

MessageDocumentHeader

MessageDocumentHeader is an energy data space specific entity with the creationDateTime of this document.

!# shows all attributes of MessageDocumentHeader.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document assembly model::MessageDocumentHeader

Order	mult.	Attribute name / Attribute type	Description
0	[0..1]	creationDateTime ESMP_DateTime	The date and time the MessageDocumenHeader was created.

##MXM##

!# shows all association ends of MessageDocumentHeader with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document assembly model::MessageDocumentHeader with other classes

Order	mult.	Class name / Role	Description
1	[0..1]	MetaInformation MetaInformation	Association Based On : CEEDS_ValidatedHistoricalData Document contextual model::MessageDocumentHeader.[] ----- CEEDS_ValidatedHistoricalData Document contextual model::MetaInformation.MetaInformation[0..1]

##MXM##

MetaInformation

MetaInformation is an energy data space specific entity referring to the information concerning the technical and functional context of the MarketDocument.

!# shows all attributes of MetaInformation.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document assembly model::MetaInformation

Order	mult.	Attribute name / Attribute type	Description
0	[0..1]	connectionId String	connectionId is an energy data space specific attribute referring to the established connection used for the communication of this document.
1	[0..1]	requestPermissionId String	The requestPermissionId is energy data space specific identification attribute holding an unique identifier issued by the energy data space to allow for tracing of the permission through serveral message exchanges. Note: the requestPermissionId is NOT the identifier issued by the Identity Service Provider as specified in the EU Implementing Act on access to Metering and Consumption Data.
2	[0..1]	dataNeedId String	dataNeedId is an energy data space specific attribute holding the type of data need requested, like â€˜last 24 hours at 1 hour intervalâ€ .
3	[0..1]	documentType String	documentType is an energy data space specific attribute referring to the type of document, comparable to the MessageTypeList in the CIM part of the message.

			Note: This is applicable if you have the need to know the content without having to open the complete document. DocumentType might contain messages not (yet) part of the MessageTypeList.
4	[0..1]	finalCustomerId String	The finalCustomerId is an energy data space specific attribute identifying the actor doing the request for data or data sharing.
5	[0..1]	dataSourceId String	dataSourceId is an energy data space specific attribute referring to the datasource from which the requested data is being queried.
6	[0..1]	defaultValues String	defaultValues is a energy data space specific attribute to hold relevant default values. The format to use can freely be chosen based on the implementation. Most commonly you would expect a semi-colon separated name value pair, like: "minPower=value; maxPower=value; avgPower=value".
7	[0..1]	regionConnector String	regionConnector is an energy data space specific attribute that specifies the regional connector being used to retrieve the data requested.
8	[0..1]	regionCountry String	regionCountry is an energy data space specific attribute that specifies the country where the regional connector is located.

##MXM##

!# shows all association ends of MetaInformation with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document assembly model::MetaInformation with other classes

Order	mult.	Class name / Role	Description
9	[0..1]	EnvAsset EnvAsset	Association Based On : CEEDS_ValidatedHistoricalData Document contextual model::MetaInformation.[] ----- CEEDS_ValidatedHistoricalData Document contextual model::EnvAsset.EnvAsset[0..1]

##MXM##

Point

The identification of the values being addressed within a specific interval of time.

!# shows all attributes of Point.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document assembly model::Point

Order	mult.	Attribute name / Attribute type	Description
0	[1..1]	position Position_Integer	A sequential value representing the relative position within a given time interval.
1	[0..1]	energyQuality_Quantity.quantity Decimal	The quantity value. The association role provides the information about what is expressed. --- The Quantity information

			associated with a given Point.
2	[0..1]	energyQuality_Quantity.quality Quality_String	The description of the quality of the quantity. --- The Quantity information associated with a given Point.
3	[0..1]	energyQuality_Quantity.type String	The description of the type of the quantity. --- The Quantity information associated with a given Point.
4	[0..1]	energyQuality_Quantity.quantityDefinition.mRID String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. --- The Quantity information associated with a given Point.
5	[0..1]	energyQuality_Quantity.quantityDefinition.quantityType String	Describes an observable property that can be measured and quantified numerically. Electricity related examples are ActivePower, ReactivePower, ActiveEnergy, ReactiveEnergy. --- The Quantity information associated with a given Point.
6	[0..1]	energyQuality_Quantity.quantityDefinition.commodityFlowOccurence CommodityDirectionKind	The direction of flow, such as that of water or electricity, for the quantity. Quantities can be for measurements, or expected future values, e.g. forecast, schedule, instruction. --- The Quantity information associated with a given Point.
7	[0..1]	energyQuality_Quantity.quantityDefinition.quantityQualifier String	Role or characteristic of the quantity type. For example: average measured value, minimum measured value,

			maximum measured value, forecast expected value, forecast lower bound, forecast upper bound, lower operational limit, upper operational limit. --- The Quantity information associated with a given Point.
8	[0..1]	energyQuality_Quantity.quantityDefinition.measurement_Unit.name MeasurementUnitKind_String	The identification of the formal code for a measurement unit (UN/ECE Recommendation 20). --- The Quantity information associated with a given Point.
9	[0..1]	energy_Quantity.quantity Decimal	The quantity value. The association role provides the information about what is expressed. --- The Quantity information associated with a given Point.
10	[0..1]	energy_Quantity.quality Quality_String	The description of the quality of the quantity. --- The Quantity information associated with a given Point.
11	[0..1]	energy_Quantity.type String	The description of the type of the quantity. --- The Quantity information associated with a given Point.
12	[0..1]	energy_Quantity.quantityDefinition.mRID String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. --- The Quantity information associated with a given Point.
13	[0..1]	energy_Quantity.quantityDefinition.quantityType String	Describes an observable property that can be measured and quantified numerically. Electricity related examples are

			ActivePower, ReactivePower, ActiveEnergy, ReactiveEnergy. --- The Quantity information associated with a given Point.
14	[0..1]	energy_Quantity.quantityDefinition.commodityFlowOccurence CommodityDirectionKind	The direction of flow, such as that of water or electricity, for the quantity. Quantities can be for measurements, or expected future values, e.g. forecast, schedule, instruction. --- The Quantity information associated with a given Point.
15	[0..1]	energy_Quantity.quantityDefinition.quantityQualifier String	Role or characteristic of the quantity type. For example: average measured value, minimum measured value, maximum measured value, forecast expected value, forecast lower bound, forecast upper bound, lower operational limit, upper operational limit. --- The Quantity information associated with a given Point.
16	[0..1]	energy_Quantity.quantityDefinition.measurement_Unit.name MeasurementUnitKind_String	The identification of the formal code for a measurement unit (UN/ECE Recommendation 20). --- The Quantity information associated with a given Point.

##MXM##

Series_Period

The identification of the period of time corresponding to a given time interval and resolution.

!# shows all attributes of Series_Period.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document assembly model::Series_Period

Order	mult.	Attribute name / Attribute type	Description
0	[1..1]	resolution Duration	The definition of the number of units of time that compose an individual step within a period.
1	[1..1]	timeInterval ESMP_DateTimeInterval	The start and end time of the period.
2	[0..1]	reason.code	The motivation of an act in coded form.

		ReasonCode_String	--- The reason information associated with a Series_Period providing motivation information.
3	[0..1]	reason.text	The textual explanation corresponding to the reason code.
		ReasonText_String	--- The reason information associated with a Series_Period providing motivation information.

##MXM##

!# shows all association ends of Series_Period with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document assembly model::Series_Period with other classes

Order	mult.	Class name / Role	Description
4	[1..*]	Point Point	The Point information associated with a given Series_Period.within a TimeSeries. Association Based On : CEEDS_ValidatedHistoricalData Document contextual model::Series_Period.[] ----- CEEDS_ValidatedHistoricalData Document contextual model::Point.Point[1..*]

##MXM##

TimeSeries

A set of time-ordered quantities being exchanged in relation to a product.

In the ESMP profile, the TimeSeries provides not only time-ordered quantities but also time-ordered information.

!# shows all attributes of TimeSeries.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document assembly model::TimeSeries

Order	mult.	Attribute name / Attribute type	Description
0	[0..1]	mRID ID_String	A unique identification of the time series. In the ESMP context, the "model authority" is defined as a party (originator of the exchange) that provides a unique identification in the context of a business exchange such as time series identification, bid identification, ... Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily

			achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
1	[1..1]	version ESMPVersion_String	The identification of the version of the time series.
2	[0..1]	businessType BusinessKind_String	The identification of the nature of the time series.
3	[0..1]	product EnergyProductKind_String	The identification of the nature of an energy product such as power, energy, reactive power, etc.
4	[1..1]	flowDirection.direction DirectionKind_String	The coded identification of the direction of energy flow. --- The flow direction associated with a TimeSeries.
5	[1..1]	dateAndOrTime.dateTime DateTime	Date and time as per ISO 8601 YYYY-MM-DDThh:mm:ss.sssZ. --- A date and/or time associated with a TimeSeries.
7	[0..1]	registeredResource.mRID ResourceID_String	The unique identification of a resource. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. --- The identification of a resource associated with a TimeSeries.

8	[0..1]	registeredResource.name String	The name is any free human readable and possibly non unique text naming the object. --- The identification of a resource associated with a TimeSeries.
9	[0..1]	registeredResource.description String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy. --- The identification of a resource associated with a TimeSeries.
10	[0..1]	registeredResource.pSRType.psrType PsrType_String	The coded type of a power system resource. --- The identification of a resource associated with a TimeSeries. --- The identification of the type of resource associated with this RegisteredResource.
11	[0..1]	registeredResource.location.mRID String	The unique identification of a location. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification, location, etc. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. --- The identification of a resource associated with a TimeSeries. --- Location of this power system resource.

12	[0..1]	registeredResource.location.name String	The name is any free human readable and possibly non unique text naming the object. --- The identification of a resource associated with a TimeSeries. --- Location of this power system resource.
13	[0..1]	registeredResource.location.geometry.identifier UUID	A universally unique object identifier. Used to uniquely identify persistent objects between CIM messages. --- The identification of a resource associated with a TimeSeries. --- Location of this power system resource. --- The geometry for a given location.
14	[0..1]	registeredResource.location.geometry.mRID String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. --- The identification of a resource associated with a TimeSeries. --- Location of this power system resource. --- The geometry for a given location.
15	[0..1]	registeredResource.location.geometry.positionPoint.xPosition String	X axis position. --- The identification of a resource associated with a TimeSeries. --- Location of this power system resource. --- The geometry for a given location.
16	[0..1]	registeredResource.location.geometry.positionPoint.yPosition	Y axis position.

		String	--- The identification of a resource associated with a TimeSeries. --- Location of this power system resource. --- The geometry for a given location.
17	[0..1]	registeredResource.location.geometry.positionPoint.zPosition String	(if applicable) Z axis position. --- The identification of a resource associated with a TimeSeries. --- Location of this power system resource. --- The geometry for a given location.
18	[0..1]	registeredResource.location.geometry.coordinateSystem.crsUrn String	A Uniform Resource Name (URN) for the coordinate reference system (crs) used to define 'Location.PositionPoints'. An example would be the European Petroleum Survey Group (EPSG) code for a coordinate reference system, defined in URN under the Open Geospatial Consortium (OGC) namespace as: urn:ogc:def:uom:EPSG::XXXX, where XXXX is an EPSG code (a full list of codes can be found at the EPSG Registry web site http://www.epsg-registry.org/). To define the coordinate system as being WGS84 (latitude, longitude) using an EPSG OGC, this attribute would be urn:ogc:def:uom:EPSG::4236. A profile should limit this code to a set of allowed URNs agreed to by all sending and receiving parties. --- The identification of a resource associated with a TimeSeries. --- Location of this power system resource. --- The geometry for a given location. --- The coordinate system used for a given geometry.
19	[0..1]	registeredResource.fuel.fuel	An indication of the fuel used

		Fuel_String	for the energy production, or part of the energy production, that is potentially fed into the grid at the related object. --- The identification of a resource associated with a TimeSeries.
20	[0..1]	reason.code ReasonCode_String	The motivation of an act in coded form. --- The reason information associated with a TimeSeries providing motivation information.
21	[0..1]	reason.text ReasonText_String	The textual explanation corresponding to the reason code. --- The reason information associated with a TimeSeries providing motivation information.
22	[0..1]	energy_Measurement_Unit.name MeasurementUnitKind_String	The identification of the formal code for a measurement unit (UN/ECE Recommendation 20). --- The unit of measure associated with the quantities in a TimeSeries.
23	[0..1]	energyQuality_Measurement_Unit.name MeasurementUnitKind_String	The identification of the formal code for a measurement unit (UN/ECE Recommendation 20). --- The unit of measure associated with the quantities in a TimeSeries.
24	[0..1]	accountingPoint.mRID MeasurementPointID_String	A unique identification of the measurement point. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this.

			For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. --- The identification of a measurement point associated with a TimeSeries.
25	[0..1]	accountingPoint.name String	The name is any free human readable and possibly non unique text naming the object. --- The identification of a measurement point associated with a TimeSeries.
26	[0..1]	accountingPoint.usagePointLocation.geoInfoReference String	(if applicable) Reference to geographical information source, often external to the utility. --- The identification of a measurement point associated with a TimeSeries. --- Location of this usage point.
27	[0..1]	accountingPoint.meterReadings.mRID ResourceID_String	The unique identification of the meter reading. In the ESMP context, the "model authority" is defined as a party (originator of the exchange) that provides an identification in the context of a business exchange. Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552 Edition 1, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. --- The identification of a measurement point associated with a TimeSeries. --- All meter readings obtained from this usage point.
28	[0..1]	accountingPoint.meterReadings.name String	The name is any free human readable and possibly non unique text naming the object.

			--- The identification of a measurement point associated with a TimeSeries. --- All meter readings obtained from this usage point.
29	[0..1]	accountingPoint.meterReadings.readings.mRID ResourceID_String	The identification of the reading. Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552 Edition 1, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. --- The identification of a measurement point associated with a TimeSeries. --- All meter readings obtained from this usage point. --- All reading values contained within this meter reading.
30	[0..1]	accountingPoint.meterReadings.readings.readingType.aggregate AggregateKind	Salient attribute of the reading data aggregated from individual endpoints. This is mainly used to define a mathematical operation carried out over 'macroPeriod', but may also be used to describe an attribute of the data when the 'macroPeriod' is not defined. --- The identification of a measurement point associated with a TimeSeries. --- All meter readings obtained from this usage point. --- All reading values contained within this meter reading. --- Type information for this reading value.
31	[0..1]	accountingPoint.meterReadings.readings.readingType.accumulation	Accumulation behaviour of a reading over time, usually

		AccumulationKind	'measuringPeriod', to be used with individual endpoints (as opposed to 'macroPeriod' and 'aggregate' that are used to describe aggregations of data from individual endpoints). --- The identification of a measurement point associated with a TimeSeries. --- All meter readings obtained from this usage point. --- All reading values contained within this meter reading. --- Type information for this reading value.
32	[0..1]	accountingPoint.meterReadings.readings.readingType.commodity CommodityKind	Commodity being measured. --- The identification of a measurement point associated with a TimeSeries. --- All meter readings obtained from this usage point. --- All reading values contained within this meter reading. --- Type information for this reading value.

##MXM##

!# shows all association ends of TimeSeries with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document assembly model::TimeSeries with other classes

Order	mult.	Class name / Role	Description
6	[0..*]	Series_Period Series_Period	The time interval and resolution for a period associated with a TimeSeries. Association Based On : CEEDS_ValidatedHistoricalData Document contextual model::TimeSeries.[] ----- CEEDS_ValidatedHistoricalData Document contextual model::Series_Period.Series_Period[0..*]

##MXM##

VHD_MarketDocument

An electronic document containing the information necessary to satisfy the requirements of a given business process.

!# shows all attributes of VHD_MarketDocument.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document assembly model::VHD_MarketDocument

Order	mult.	Attribute name / Attribute type	Description
0	[1..1]	mRID ID_String	The unique identification of the document being exchanged within a business process flow. In the ESMP context, the "model authority" is defined as a party (originator of the exchange) that provides an identification in the context of a business exchange such as document identification, ... Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
1	[1..1]	createdDateTime ESMP_DateTime	The date and time of the creation of the document.
2	[0..1]	description String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.
3	[0..1]	revisionNumber ESMPVersion_String	The identification of the version that distinguishes one evolution of a document from another.
4	[0..1]	type MessageKind_String	The coded type of a document. The document type describes the principal characteristic of the document.
6	[1..1]	process.processType ProcessKind_String	The identification of the nature of process that the document addresses. --- The Process associated with an electronic document header that is valid for the whole document.
7	[1..1]	receiver_MarketParticipant.mRID PartyID_String	The identification of a party in the energy market. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
8	[0..1]	receiver_MarketParticipant.name String	The name is any free human readable and possibly non unique text naming the object.
9	[1..1]	receiver_MarketParticipant.marketRole.type MarketRoleKind_String	The identification of the role played by a market player.
10	[1..1]	sender_MarketParticipant.mRID	The identification of a party in the energy market.

		PartyID_String	In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
11	[0..1]	sender_MarketParticipant.name String	The name is any free human readable and possibly non unique text naming the object.
12	[1..1]	sender_MarketParticipant.marketRole.type MarketRoleKind_String	The identification of the role played by a market player.

##MXM##

!# shows all association ends of VHD_MarketDocument with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document assembly model::VHD_MarketDocument with other classes

Order	mult.	Class name / Role	Description
5	[1..*]	TimeSeries TimeSeries	The time series that is associated with an electronic document. Association Based On : CEEDS_ValidatedHistoricalData Document contextual model::VHD_MarketDocument.[] ----- CEEDS_ValidatedHistoricalData Document contextual model::TimeSeries.TimeSeries[1..*]

##MXM##

Datatypes

The list of datatypes used for the CEEDS_ValidatedHistoricalData Document assembly model is as follows:

- ESMP_DateTimeInterval compound
- BusinessKind_String datatype, codelist BusinessTypeList
- DirectionKind_String datatype, codelist DirectionTypeList
- EnergyProductKind_String datatype, codelist EnergyProductTypeList
- ESMP_DateTime datatype
- ESMPVersion_String datatype
- Fuel_String datatype, codelist FuelTypeList
- ID_String datatype
- MarketRoleKind_String datatype, codelist RoleTypeList
- MeasurementPointID_String datatype, codelist CodingSchemeTypeList
- MeasurementUnitKind_String datatype, codelist UnitOfMeasureTypeList
- MessageKind_String datatype, codelist MessageTypeList
- PartyID_String datatype, codelist CodingSchemeTypeList
- Position_Integer datatype

- ProcessKind_String datatype, codelist ProcessTypeList
- PsrType_String datatype, codelist AssetTypeList
- Quality_String datatype, codelist QualityTypeList
- ReasonCode_String datatype, codelist ReasonCodeTypeList
- ReasonText_String datatype
- ResourceID_String datatype, codelist CodingSchemeTypeList
- YMDHM_DateTime datatype